Operational Identification and Calibration in Two-Mode Quasifree Systems

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Formal statements

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Abstract. We separate three logically different problems: determining a finite-mode state from measured observables, determining parameters in an explicitly selected thermal model, and deciding whether that model adequately describes observations. A two-mode gauge-invariant state admits an exact quasifree unhalved trace-norm witness and sharp occupation-only feasibility bounds. Restricted complete joint-number histograms identify Wilson–Dirac thermal parameters under a shared in-plane detector tilt, whereas unrestricted detector orientations admit aliases. Four momentum settings identify a first-harmonic spectrum from orbital-axis-invariant total-number data but suffer two singular differential strata. One calibrated interior orbital statistic restores first-order rank. Statistical guarantees are conditional on calibration and independent preparation, not consequences of exact algebra alone.

Keywords: CAR algebra; quasifree states; identifiability; calibration; quantum statistics.

1. Observation objects and the exact two-mode witness

We use the standard gauge-invariant CAR conventions [1]. Simultaneous sample bounds below employ Hoeffding's concentration inequality [2].

Fix two fermionic modes a_1, a_2 and a gauge-invariant state ρ commuting with the total-number operator. Let q_{ab}, $a, b \in \{0, 1\}$, be the joint-number probabilities, and let $C_{ij} = \langle a_i^\dagger a_j \rangle_\rho$ be its one-particle covariance. The state has a one-dimensional vacuum block, a positive 2×2 one-particle block, and a one-dimensional double-occupation block. No quasifree assumption has yet been made. This article uses the occupation-index convention $C_{ij} = \langle a_i^\dagger a_j \rangle$. The operator-index convention $C^{\mathrm{op}}_{ij} = \langle a_j^\dagger a_i \rangle$ used in *Restricted Observables and CAR-State Distinguishability* satisfies $C^{\mathrm{op}} = C^{\mathsf{T}} = \overline{C}$; spectra, determinants, and $|C_{12}|^2$ are unchanged.

For the present model, $\|X\|_1 = \mathrm{Tr}\,|X|$ denotes the **unhalved Schatten trace norm**. The conventional operational trace distance is $T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$; numerical formulae below use the unhalved convention unless T is written explicitly.

Let ω_C be the *gauge-invariant quasifree* state with the same covariance. Define

$$W = q_{10}q_{01} - q_{00}q_{11} - |C_{12}|^2.$$

Theorem 1 (two-mode quasifree distance identity). Exactly for the stated two-mode gauge-invariant class,

$$\boxed{\|\rho - \omega_C\|_1 = 4|W|, \qquad W = 0 \Longleftrightarrow \rho = \omega_C.}$$

Proof. Set $x = C_{11}$ and $y = C_{22}$. The number probabilities determine $x = q_{10} + q_{11}$, $y = q_{01} + q_{11}$. The quasifree Wick formula sets $q_{11}^G = xy - |C_{12}|^2$, giving $W = q_{11}^G - q_{11}$. The two states have the same one-particle coherence (up to the fixed CAR basis convention), so their difference has four diagonal entries $W, -W, -W, W$ in occupation-number sectors. The eigenvalues of this difference are those four numbers, hence the trace norm equals $4|W|$, while $T(\rho, \omega_C) = 2|W|$. □

An experiment that records the joint histogram **alone** has no direct access to $|C_{12}|^2$ for arbitrary states: measuring covariance quadratures in a consistently calibrated orbital basis is a separate requirement. In particular, dephasing

a genuinely coherent Gibbs state in the counting basis can preserve its number histogram while changing C_{12}. The witness must not be called a certification of a particular Gibbs Hamiltonian; it certifies quasifreeness only in the specified state class.

For probability vectors $p, q \in \mathbb{R}^4$ on the simplex, let $F(q) = q_{10}q_{01} - q_{00}q_{11}$. The derivative has ℓ^1 norm $\sum_i |\partial_i F| = 1$, yielding

$$\boxed{|F(p) - F(q)| \le \|p - q\|_\infty.}$$

This sharpens the histogram contribution to a simultaneous finite-shot witness certificate. Readout bias, the two coherence quadratures and systematic basis drift must be bounded separately. An empty model-relative confidence set is a valid conditional rejection; a nonempty one is not proof that the physical state belongs to the chosen family.

2. Explicit thermal family and complete histogram reconstruction

Let $a = m + 1 \ge 1$, $\lambda > 0$, $|\epsilon| < 1$, $\eta \in \mathbb{R}$ and

$$E_a(k) = \sqrt{\sin^2 k + (a - \cos k)^2}, \qquad D(k) = \lambda(1 + \epsilon \cos k)E_a(k).$$

For one common, unknown **signed in-plane detector tilt** $\theta \in (-\pi/2, \pi/2)$, set $r_\theta(k) = \{(a - \cos k)\cos\theta + \sin k \sin\theta\}/E_a(k)$. The full joint-number histogram, ordered $(00, 10, 01, 11)$, is

$$q(k) = Z^{-1}(1, e^\eta[\cosh D - r_\theta \sinh D], e^\eta[\cosh D + r_\theta \sinh D], e^{2\eta}),$$
$$Z = 1 + 2e^\eta \cosh D + e^{2\eta}.$$

For strict positive observations the algebraic inverse is

$$\eta = \tfrac{1}{2}\log(q_{11}/q_{00}), \quad D = \operatorname{arcosh}\frac{q_{10} + q_{01}}{2\sqrt{q_{00}q_{11}}}, \quad r_\theta = \frac{q_{01} - q_{10}}{2\sqrt{q_{00}q_{11}}\sinh D}.$$

Theorem 2 (three-momentum identification, conditional). For any $k_* \in (\pi/2, \pi)$, complete histograms at $(\pi/2, k_*, \pi)$ identify $(m, \lambda, \epsilon, \eta, \theta)$ uniquely *within the above shared-in-plane-tilt model*. They have a full-column-rank parameter Jacobian for every finite admissible tuple.

Proof. Write $k_1 = \pi/2$, $k_2 = k_*$ and $\phi_a(k) = \arctan[\sin k/(a - \cos k)]$. Then $r_\theta(k) = \cos(\phi_a(k) - \theta)$ at the two interior nodes. The angles lie in $(0, \pi/4]$, decrease strictly with a, and decrease strictly with k on the stated interval. The observation at π gives $\cos\theta$, so a possible alias has $\theta' = \theta$ or $\theta' = -\theta$.

Suppose first that $\theta' = \theta$. If $a' \ne a$, strict monotonicity in a rules out equal interior angles. Equality of their cosines must therefore use the reflected branch at both nodes: $\phi_a(k_i) + \phi_{a'}(k_i) = 2\theta$ for $i = 1, 2$. The left side decreases strictly from k_1 to k_2, a contradiction. Thus $a' = a$ in this case.

If $\theta' = -\theta$, the reflected cosine branch would require $\phi_a(k_i) + \phi_{a'}(k_i) = 0$, which is impossible. The remaining branch requires $\phi_a(k_i) - \phi_{a'}(k_i) = 2\theta$ at both nodes. For $a \ne a'$, its tangent is

$$(a' - a)\sin k/[aa' + 1 - (a + a')\cos k],$$

whose absolute value is strictly smaller at k_2 than at k_1. This again is impossible. If $a = a'$, the equation forces $\theta = 0$. The angle ranges exclude extra 2π branches. Hence a, θ are unique. Two splittings at distinct cosine nodes determine $(\lambda, \lambda\epsilon)$ by an invertible linear system, and the vacuum/double-occupation ratio determines η.

For first-order rank, put $p_i = \partial_a \phi_a(k_i) = -\sin k_i/E_a(k_i)^2$ and $s_i = \sin(\phi_a(k_i) - \theta)$. The rows of the directional Jacobian in (a, θ) are $(-s_i p_i, s_i)$ and $(0, -\sin\theta)$ at π. If $\theta \ne 0$, at most one s_i vanishes, since the two interior

angles differ. The row at π and a nonzero interior row therefore have nonzero determinant. At $\theta = 0$, the determinant from the two interior rows is

$$\sin\phi_a(k_1)\sin\phi_a(k_2)(p_2 - p_1) > 0,$$

because $p_2 > p_1 = -1/(a^2+1)$. The splitting map has full rank in (λ, ϵ) since $\lambda > 0$ and the cosine nodes differ; η supplies the remaining independent coordinate. The histogram inversion is smooth at these nodes because all probabilities and splittings are strictly positive. This proves full column rank. □

3. Four total-number momenta: exact identification versus conditioning

The total-number categories eliminate the orbital direction:

$$(p_0, p_1, p_2) = Z^{-1}(1, 2e^{\eta}\cosh D, e^{2\eta}).$$

Hence four distinct, known cosine nodes determine four samples of $Q(c) = D(c)^2$, a cubic for the first-harmonic family. In the branch $a = m + 1 \geq 1$, its exact polynomial determines the Wilson mass and thermal parameters under the positivity assumptions. The four-node observation map is globally injective in this specified family.

It is not everywhere a *regular* inverse. Put

$$r = \frac{1}{2}(a + a^{-1}), \quad L = \lambda\sqrt{a}, \quad Q(c) = 2L^2(1+\epsilon c)^2(r - c).$$

Two separate differential singularities occur: $a = 1$ (the derivative dr/da vanishes), and the positive-mass root collision $1 + \epsilon r = 0$ (the cubic-coefficient Jacobian drops rank). At either, there exist admissible paths with mass separation of order h but total-number probability separation of order h^2, yielding $D_{\mathrm{KL}} = O(h^4)$ for fixed finite designs. Thus $N^{-1/4}$ lower-bound behavior is possible for total-number-only mass estimation near these strata; it is not the same claim as globally nonunique parameters.

Theorem 3 (calibrated directional rank recovery). One additional orbital-direction observable in the *known Hamiltonian basis* at an interior k has

$$s(a,k) = \frac{a - \cos k}{E_a(k)}, \qquad \boxed{\partial_a s(a,k) = \frac{\sin^2 k}{E_a(k)^3} > 0.}$$

This is nonzero in the mass directions responsible for both singularities, restoring full-column first-order rank for the augmented ideal observation map. It may be measured at an existing momentum. The assumption of independently calibrated and stable orbital axes is essential; this is not a claim that such calibration is experimentally available.

Explicit Jacobian minors. Set $Q(c) = \lambda^2(1+\epsilon c)^2(a^2 + 1 - 2ac)$ and write its coefficients as $Q_0, \ldots, Q_3$. The four 3×3 minors of $\partial(Q_0, Q_1, Q_2, Q_3)/\partial(a, \lambda, \epsilon)$, in lexicographically ordered row triples, are exactly

$$8\lambda^5(a^2-1)[\epsilon(a^2+1) + 2a]\,(1, \epsilon, \epsilon^2, \epsilon^3).$$

Both singular strata follow. Independently, $\det \partial(Q_0, Q_1)/\partial(\lambda, \epsilon) = 4\lambda^3(a^2+1)^2 > 0$. Adding the calibrated interior directional statistic contributes a maximal minor with absolute value

$$\frac{4\lambda^3(a^2+1)^2\sin^2 k}{(a^2 + 1 - 2a\cos k)^{3/2}} > 0.$$

This proves **first-order local rank recovery**, not a finite-sample confidence guarantee or calibration feasibility.

4. Simultaneous finite-error inference

At three momenta, let $\hat{q}_{i,s}$ be multinomial empirical fractions from n_i independent preparations. Suppose every setting has a justified deterministic per-category readout error β_i. Hoeffding's union bound yields with probability at least $1-\alpha$,

$$|\hat{q}_{i,s} - q_{i,s}(\xi_0)| \le \sqrt{\frac{\log(24/\alpha)}{2n_i}} + \beta_i \quad \text{for all 12 cells.}$$

The compatible set of parameters in a predeclared compact admissible domain is therefore a valid model-relative confidence region. Converting this into a diameter or sample guarantee additionally requires a quantitative inverse modulus and calibration bounds. Profiled Fisher information describes only the local tangent experiment, not global aliasing or goodness of fit.

5. Exact histogram-only quasifree feasibility and partial identification

Theorem 4 (sharp counting-data theorem). Let $q = (q_{00}, q_{10}, q_{01}, q_{11})$ be any nonnegative normalized joint-number histogram. Consider *all* gauge-invariant two-mode density operators producing precisely q, with their individually covariance-matched quasifree states ω_C. Put $a = q_{10}q_{01}$ and $b = q_{00}q_{11}$. Then

$$\boxed{\min_{\rho\mapsto q} \|\rho - \omega_{C(\rho)}\|_1 = 4(b-a)_+, \qquad \max_{\rho\mapsto q} \|\rho - \omega_{C(\rho)}\|_1 = 4\max\{b, |a-b|\}.}$$

Every value between the displayed extrema is attained. There exists a gauge-invariant quasifree state producing q **if and only if** $a \ge b$. Every gauge-invariant state producing q is quasifree **if and only if** $a = b = 0$.

Proof. In the occupation basis $|00\rangle, |10\rangle, |01\rangle, |11\rangle$, every state with histogram q has a block-diagonal density matrix with one-particle block

$$\begin{pmatrix} q_{10} & z \\ \bar{z} & q_{01} \end{pmatrix}.$$

Its positivity is equivalent to $0 \le |z|^2 \le a$. By Theorem 1, $\|\rho - \omega_C\|_1 = 4|a - b - |z|^2|$. As $u = |z|^2$ ranges continuously over $[0, a]$, the signed defect ranges over $[-b, a-b]$. Elementary optimization of its absolute value gives both sharp extrema. Zero is attainable exactly when $a \ge b$. The distance vanishes for *every* permissible coherence only when $a = 0 = b$. □

Finite-error falsification. If a normalized empirical histogram $\hat{q}$ obeys the simultaneous deterministic error envelope $\|q - \hat{q}\|_\infty \le \delta$, the elementary product Lipschitz bound for $F(q) = q_{10}q_{01} - q_{00}q_{11}$ gives

$$\boxed{\|\rho - \omega_C\|_1 \ge 4[\hat{q}_{00}\hat{q}_{11} - \hat{q}_{10}\hat{q}_{01} - \delta]_+.}$$

Thus counting-only data can directly *reject* gauge-invariant quasifreeness when the product of empty and double-occupation probabilities is too large. The reverse inequality does not certify quasifreeness without measured coherence, except for the degenerate $a = b = 0$ cases. Finite-shot confidence statements require a justified simultaneous error envelope; the deterministic theorem itself does not assume a sampling model.

Exact examples

- $q = (9/20, 1/20, 1/20, 9/20)$ forces **unhalved trace-norm distance** at least $4/5$ (operational trace distance at least $2/5$), regardless of any unmeasured coherence.
- $q = (1/10, 2/5, 2/5, 1/10)$ admits both a quasifree state and non-quasifree states, with **unhalved trace-norm distance** ranging from zero to $3/5$ (operational trace distance from zero to $3/10$).
- The vacuum histogram $q = (1, 0, 0, 0)$ admits only a quasifree state.

These are feasibility and falsification results on a specified two-mode state class, not a certification of the Gibbs Hamiltonian or an experimental apparatus.

6. Observation-alias controls and what counting alone cannot prove

For the exact histogram $q = (1/10, 2/5, 2/5, 1/10)$, the one-particle coherence z can satisfy $0 \le |z|^2 \le 4/25$. By Theorem 1 the covariance-matched quasifree distance is $4|3/20 - |z|^2|$. Thus $|z|^2 = 3/20$ is quasifree and $z = 0$ is nonquasifree (distance $3/5$), **with exactly the same joint counting data**. A fully general stochastic readout map can likewise send every underlying histogram to one common observed histogram. These are exact aliases, not noise effects; calibrating the observation map and measuring coherence are logically separate premises.

For the simultaneous 12-cell Hoeffding envelope the independence condition is *independent experimental preparations within each setting*; it is not an assertion of independent categories within a single multinomial draw. Readout systematic error must be bounded separately. Such a confidence region is conditional on the specified data-generating model and does not prove its truth.

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